

Title: Automated Diabetic Retinopathy Classification Using Texture Feature Extraction and K-Nearest Neighbors (KNN)

Abstract

Diabetic Retinopathy (DR) is a severe medical condition caused by microvascular damage to the retina. Manual diagnosis requires expert ophthalmologists and is highly time-consuming. This project proposes an automated framework to detect DR from retinal fundus images using Gray-Level Co-occurrence Matrix (GLCM) for texture feature extraction and a K-Nearest Neighbors (KNN) classifier. The system preprocesses raw images, extracts high-dimensional statistical texture properties, and classifies images into binary stages: Normal (Class 0) and Diabetic Retinopathy (Class 1).

1. Introduction

Diabetic Retinopathy is one of the leading causes of blindness globally. Early screening can save a patient's vision, but manual diagnostic infrastructure is severely limited in rural and developing areas. Digital image processing and machine learning offer a scalable solution. This study focuses on leveraging surface textures of the retina, such as microaneurysms, hemorrhages, and exudates, which significantly alter the statistical pixel distribution compared to healthy eyes.

2. System Architecture & Methodology

The pipeline is divided into three major architectural layers:

A. Dataset Cleaning & Structural Validation

- The raw repository is dynamically verified via `DatasetCleaner`.
- Corrupted, unreadable, or missing data files are systematically isolated before the training cycle to prevent pipeline failure.

B. Feature Extraction (GLCM Processing)

Instead of feeding raw pixel values into the machine learning model, which creates huge spatial complexity, the project employs Scikit-Image's texture descriptors:

- Images are converted to grayscale and normalized.
- Graycomatrix computes the spatial relationship between neighboring pixels across multiple angles (0° , 45° , 90° , 135°).
- Graycoprops extracts explicit statistical parameters including:
 1. *Contrast* (Local intensity variations)
 2. *Correlation* (Linear dependence of gray levels)
 3. *Energy* (Uniformity of texture mapping)
 4. *Homogeneity* (Closeness of element distribution)

C. Classification Layer (KNN)

- The extracted vectors are passed through a `StandardScaler` to remove biometric scaling bias.
 - A hyperparameter-tuned K-Nearest Neighbors (KNN) model computes the spatial Euclidean distance between feature points to predict the diagnostic class.
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3. Experimental Pipeline & Setup

- Language: Python 3.14
 - Libraries: OpenCV (Image normalization), Scikit-Learn (KNN architecture & standard scaling), Matplotlib & Seaborn (Analytical visualizations), Joblib (Model serialization).
 - Data Split: 80% Training Data, 20% Production Validation Data.
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4. Evaluation Metrics

The model's validation performance is verified via standard classification evaluation parameters:

- Precision & Recall: To track false positives and ensure no infected eyes are misclassified as healthy.

- Confusion Matrix: Automatically rendered and exported via `Seaborn Heatmaps` to `analytics_outputs/confusion_matrix.png` for transparency.
 - ROC-AUC Score: To evaluate the model's capability to distinguish between the target disease boundaries.
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5. Deployment Strategy

The finalized pipelines are exported to structural serialized binary payloads (`.pkl`) inside the `production_models/` directory:

1. `dataset_standard_scaler.pkl`: Ensures new test images undergo identical mathematical scaling.
2. `optimized_retinopathy_knn.pkl`: The frozen production model ready for deployment or integration into a web app.